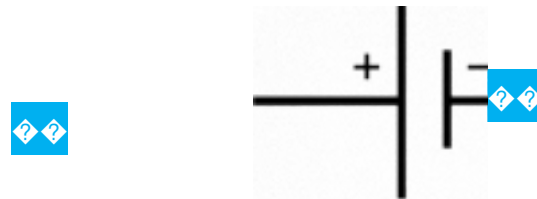
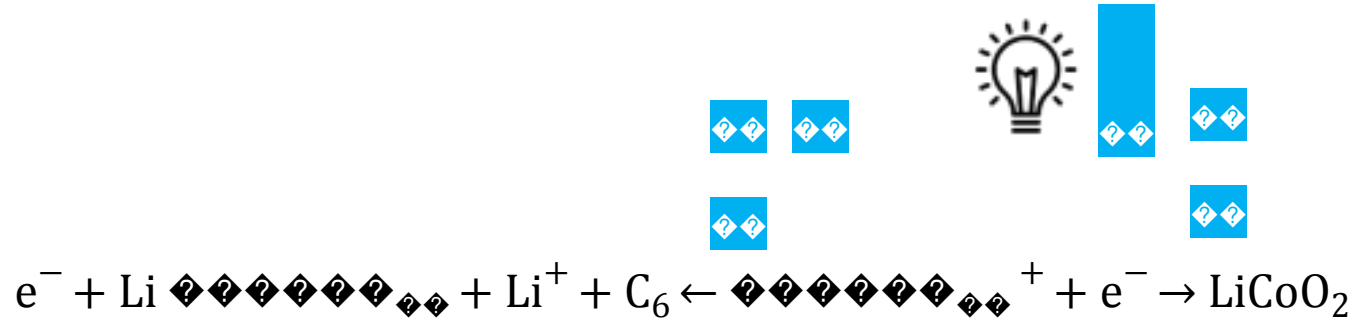


Batteries

Device used for storing electrochemical energy

The elementary element is the cell.

The principle consists in exploiting oxidation reaction at the negative pole «**anode**», generating an excess of electrons and a reduction reaction at the positive pole «**cathode**» where the electrons are «absorbed». The electron flow (**current**) that can supply a load.





M
E

M
B
R

A

N

6

kineton

E

Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018



Batteries Main characteristics

- **Open Circuit voltage (Open circuit Voltage):** Voltage between terminals in **ZERO CURRENT CONDITIONS**.
- **(charge) Maximum Capacity:** Maximum amount of electrical charge stored (Ah).
- **Mass Energy Density:** Is the amount of energy that can be stored in a battery per mass unit (kJ/Kg, Wh/Kg)Is a characteristic of the battery/chemistry. The higher, the better.
- **Power Density:** Power per unit of mass that can be delivered.
- **Cycle Life:** defined as the number of **STANDARD** cycles that the battery can undergo before its capacity falls below 80%.
- **Current rate (C-rate):** measure of the rate at which a battery is charged or discharged relative to its capacity (a 500mAh battery, being discharged at 5000mA, has a C-rate of 10C).

○ **Charge:** current amount of electrical charge that can be used to generate power (Ah, mAh) ○

State of Charge (SOC): ratio between available capacity and maximum capacity: $\frac{\text{SOC}}{\text{SOC}_{\text{max}}} \in [0\%, 100\%]$.

○ **Nominal Voltage:** Voltage at 50% SOC during a discharge cycle at 0.2C.

Power density	<2.000W/kg	180-200W/kg
Cycle life	2.500-12.000	50-100

Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018

Kineton

Batteries Some figures

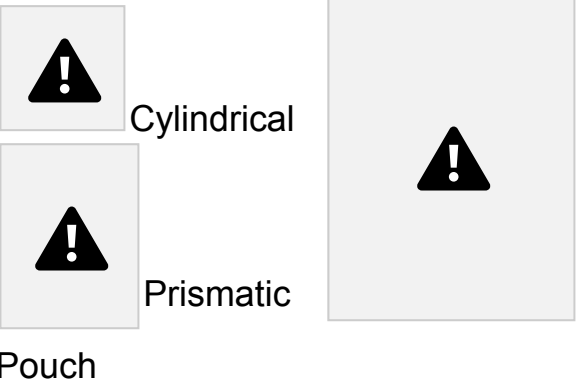


Typical battery types:

Gasoline:
44.000kJ/kg
Diesel fuel:
40.000kJ/kg

(cell) open circuit voltage	2.5 – 4.3V	1.75
Mass energy density	>500kJ/kg	140k

A typical engine: $\sim \frac{\text{Power}}{\text{Energy}}$

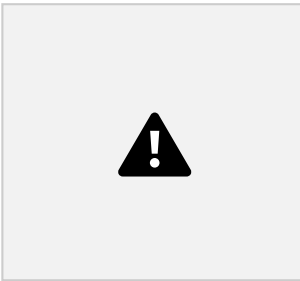


Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018

Cells (or batteries) can be connected in **series** or **parallel** in order to form **battery packs** of the desired capacity and voltage.

Batteries Automotive

Lead-acid batteries



- Used as auxiliary battery in vehicles (12-24V).
- Can be kept at very high state of charge (100%) without any issue.
- Low cost, high reliability.
- Can provide bursts of extremely high current without risk of fire/explosion (thousands of Amps).
- Resists to high temperatures/shocks.
- >99% recyclable.

Lithium batteries

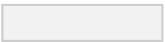
- Used as traction batteries.

- Extremely high energy and power density.
- Very expensive.
- Chance of explosion if overheated, subjected to shock, short circuited.



Ni-mh batteries

- .Used as traction batteries in some hybrid powertrains (Toyota Prius until 2015)
- Inexpensive, safe, reliable.
- Lower energy density compared to Li-based batteries.
- Subject to mild (reversible) lazy battery effect.



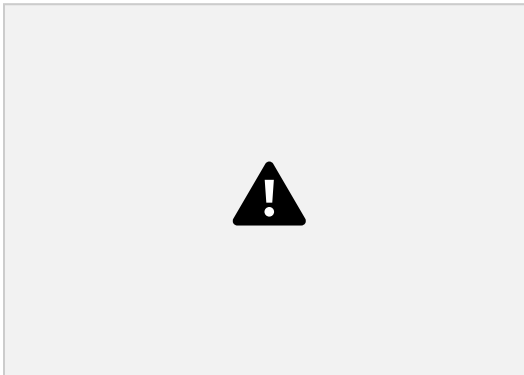
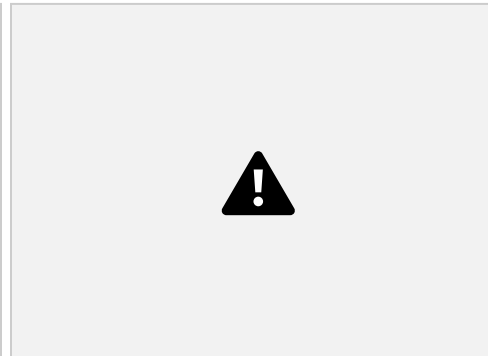
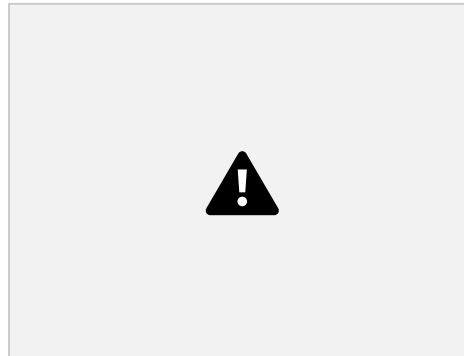
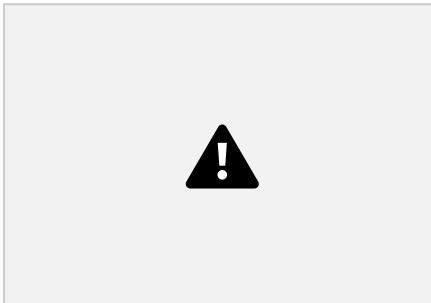


Batteries Discharge curves

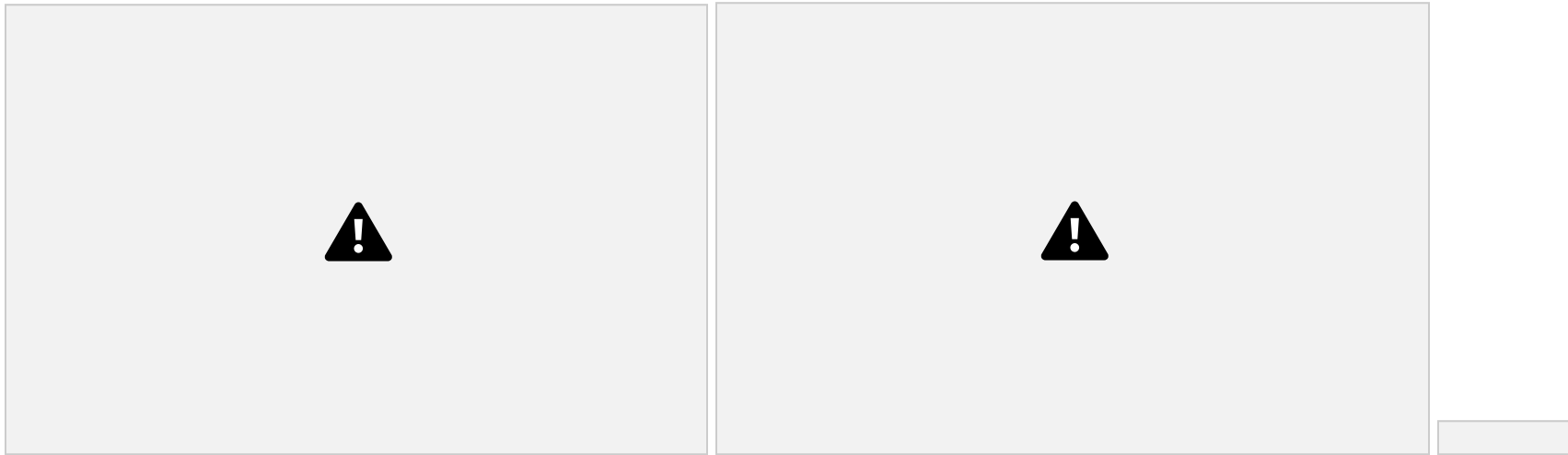
1) The voltage at the battery terminals **lowers as the discharge current (C-rate) increases**. 2) The area below the curve (harvested energy) reduces with increasing current. 3) Not all batteries behave in the same way, battery design can be optimized for maximum capacity (top chart) or maximum available current (bottom chart).

Statements 1 and 3 suggest the **simplest battery model: an ideal generator with a resistor in series (ESR)**.

Both η_{coulomb} and η_{voltage} depend on **SOC and temperature**.

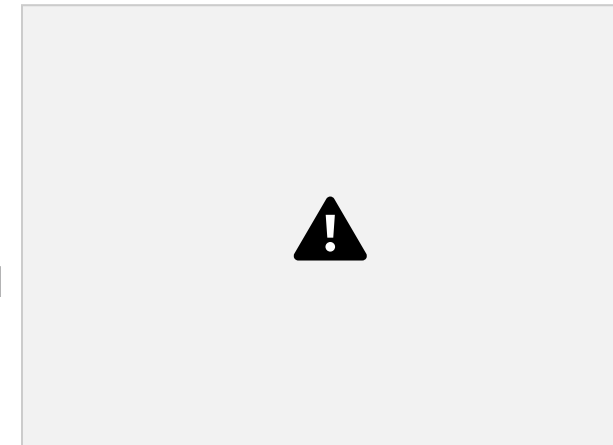


Battery actual capacity is strongly affected by operating temperature as well



Batteries Aging

- Another factor that strongly affects the battery behaviour is the number of cycles that the battery has underwent (**ageing**). An aged battery shows a higher internal resistance (top left)
- The internal resistance affects the battery self-heating during operation, aged batteries produce more heat in both charging and discharging.
- Aged batteries are subject to higher voltage drop when discharged (at constant C-rate) and higher voltage raise when charged (at constant C-rate)
- Battery capacity reduces dramatically as the number of cycles increases (left bottom)



- The charge-discharge bandwidth affects the battery aging rate (left bottom).
- Temperature affects the battery aging rate as well (right bottom): high temperatures increase battery degradation.
- The battery degradation strongly increases if the battery is **charged** at high currents (high voltages, top-right).

Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018

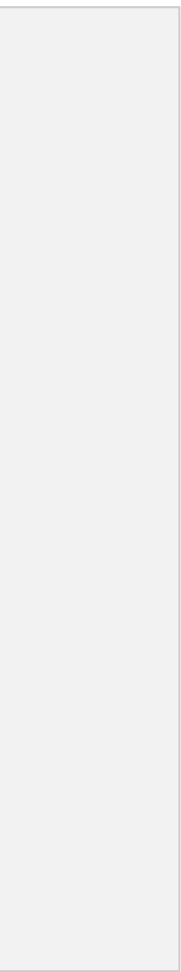


- Batteries shall operate at room temperature (20-30°C) in order to suffer the smallest degradation possible and deliver optimal performance

- **Too cold:** reduced capacity, reduced drainable/injectable current (high ESR).
 - **Too hot:** high degradation but high performance. **Temperatures above 55°C are dangerous.**
- The battery system shall be designed in order to limit current:
 - Reduce ohmic losses
 - Reduce heating (and thus, degradation)
 - Exploit the maximum battery capacity
 - Reduce battery degradation during charge
 - ESR estimation can give indication about battery degradation
 - Since battery temperature has a significant impact on battery operation (and due to safety concerns), the temperature and temperature gradients inside battery packs have to be controlled. Same for temperature of different battery packs (in case of several battery packs).

Series Parallel $\diamond\diamond = \diamond\diamond_{\diamond\diamond} = \diamond\diamond_{\diamond\diamond} = \diamond\diamond_{\diamond\diamond}$





In order to keep the ohmic losses low, higher voltage levels are
desirable for power systems.

??₁

+

??₁

??₁

Battery packs are normally formed by several batteries connected in series in order to increase the output voltage (up to 800V in some applications).

The capacity and maximum drainable current of a string of batteries

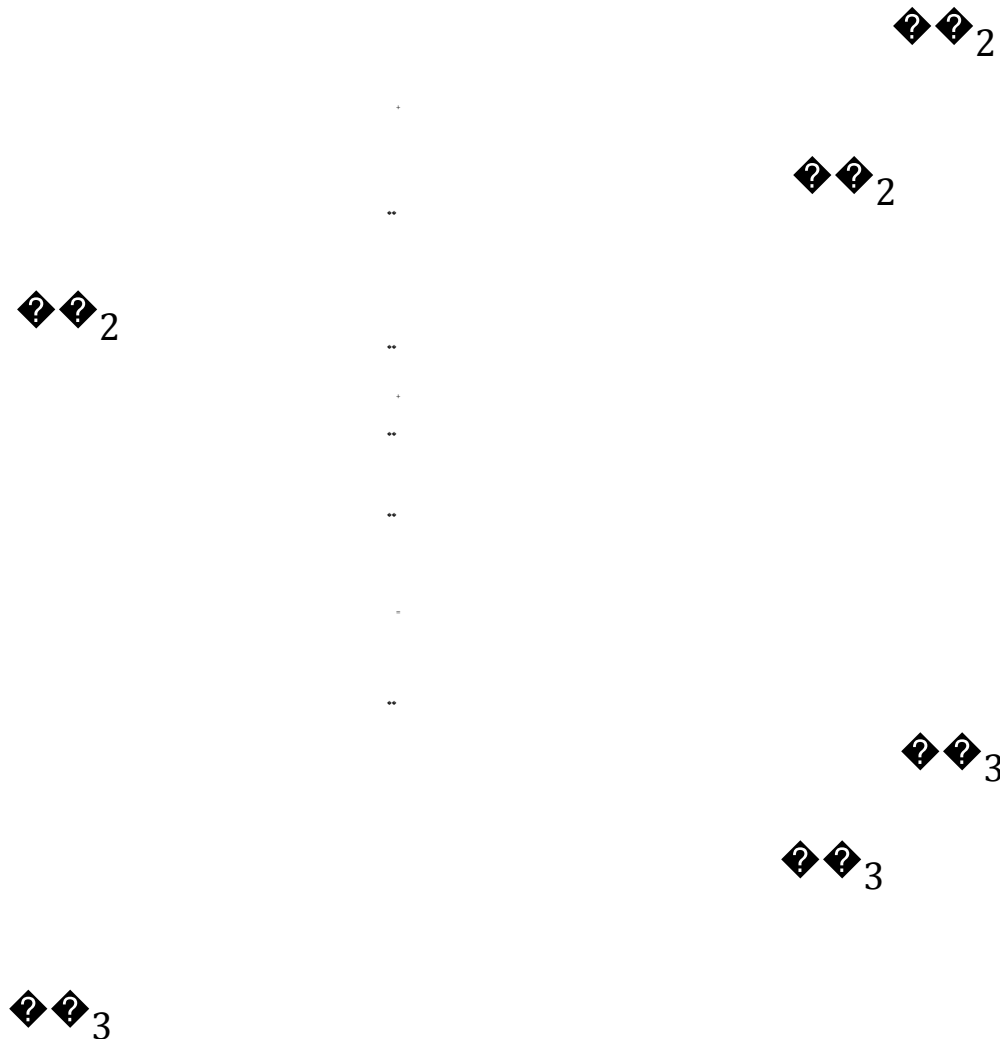
in **series** is ideally the same of the single battery.

Elements in series share the same current.

Capacity and drainable current can be limited by connecting batteries (or strings of series-connected batteries) **in parallel**.

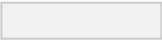
Elements in parallel share the same voltage.

Elements in parallel can virtually be decoupled if the main



constraint of the power system is the supply voltage.

$$- + \begin{matrix} \diamond \diamond \end{matrix} = \begin{matrix} \diamond \diamond \end{matrix} \begin{matrix} \diamond \diamond \end{matrix} = \begin{matrix} \diamond \diamond \end{matrix} \begin{matrix} \diamond \diamond \end{matrix} = \begin{matrix} \diamond \diamond \end{matrix} \begin{matrix} \diamond \diamond \end{matrix}$$



Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018



Batteries Series

In case of a more degraded battery (the middle one in figure), or with a lower SOC, or just with a lower temperature in a **string in series**, the common current flowing through all the batteries **will cause a higher battery heating**.

At high currents, if the ESR is very high,

undervoltage on the weak battery may occur.
Analogously, during charge, overvoltage on the weak battery may occur.

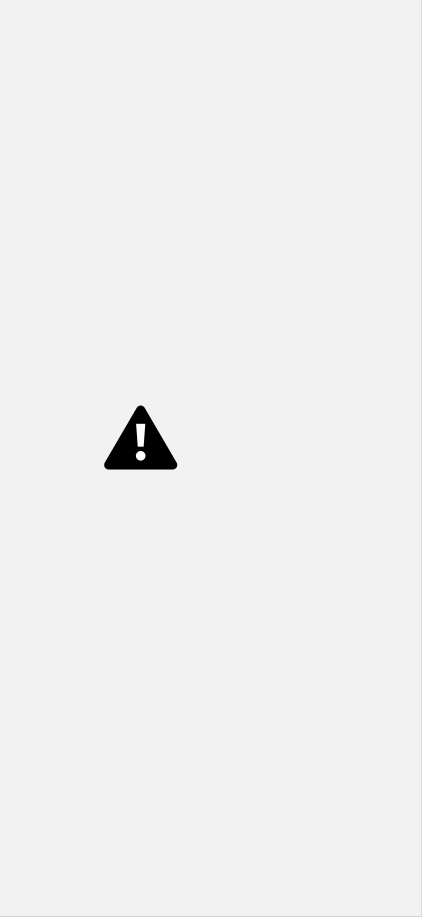
If the internal resistance of the battery is significantly higher, this can limit the maximum drainable/injectable

Discharge $\diamond \diamond$

current.

In a battery pack with many cells in series, one faulty cell can jeopardize the usability of the whole battery pack!

Charge $\diamond \diamond$



??
??
??
??
??
1 = ??

??
??
??
1

??
??
??
??
??
2 = ??

??
??
??
2

??
??
??
??
??
3 = ??

??
??
??
3 -

+

??
1

??
2

??
3



??
??
??
??
??
1 = ??

??
??
??
1

??
??
??
??
??
2 = ??

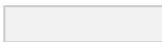
??
??
??
2

??
??
??
??
??
3 = ??



+

-



Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018



Batteries Parallel

In the case of some Battery Electric Vehicles, there may be two or more battery packs connected in parallel

on a HV bus through **high voltage contactors (high voltage relays)**.

In case of a battery with a lower SOC, or just with a lower temperature (both causing a lower Voc) in a **parallel**, the batteries with a higher voltage force a current towards the lower voltage battery

with no load on the HV bus as the contactors are closed (backcurrent).

During charge, the increased voltage difference between the HV bus and the battery pack can force several amps on the weak battery pack (overcurrent).

During discharge, the current provided by the weak battery pack is lower than the healthy battery packs, that can undergo overload.

??
3
+

Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018

Discharge ??

Real case:

- 3 BPs made by 108 cells in series (18650, max cont. Disch. curr. 20 A , max cont. charge current 10A).
- ~356 – 453 output voltage (0-100% SOC) • $3.7\Omega \div 4.5\Omega$ overall ESR.

??
1

??

??
2

??



Batteries Temperature control

- The temperature of the batteries shall be kept between certain bounds to assure efficient, safe and long operation.

- The **temperature gradients** inside a battery pack and the **temperature differences** between different battery packs shall be minimum.
- Batteries in BEV (battery electric vehicles) and some PHEV (plugin hybrid electric vehicles) require a water cooling system similar to that of conventional vehicles.
- The desired temperature level is lower, so **additional cooling** through an active refrigeration system may be necessary (like A/C system).
- Extremely low temperatures may cause the unusability of lithium batteries or the impossibility to recharge them. **Battery heating system is required.**
- **Active cooling and heating require energy.**
- **In some applications, the battery cooling shares the cooling system with the cabin A/C system**



The Battery Management System is the ECU delegated to managing:

- Battery (or batteries) connection/disconnection to the HV bus.
- **Battery balancing**
- Thermal management
 - Cooling request (the ECU managing the cooling system is separate and could be on a separate network)
- Heating request (in BEVs the battery heaters are normally directly controlled by the BMS).
- Battery monitoring
 - SOC estimation (open circuit voltage only gives a vague indication on the actual SOC).
 - SOH (state of health) estimation (many parameters can be monitored along with ESR, described my more

accurate models).

- Maximum **CHARGE**/discharge currents.
- Battery diagnosis
 - Over/undervoltage
 - Overcurrent
 - Overtemperature
 - **Insulation measurement**

Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018



BEV

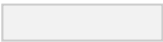


No emissions (locally)

Large battery required for acceptable range (application dependent)

No waste heat available (heating have to be actively provided).

Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018





REEV/SERIES HYBRID

Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018



Fixed point thermal engine operation (and battery charger) – MAXIMUM EFFICIENCY

Maximum energy recuperation (thanks to the supercapacitor)– MAXIMUM EFFICIENCY

Reduced battery size (depends if the thermal engine is small – REEV or large – SHEV).

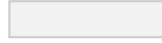
Duplicated power electronics (electric drives, generator and motor)

Almost all ICE components are present (tank, auxiliaries, cooling circuit) along with the e-PWT components



Hybrid powertrains

- Coupling a conventional powertrain with a battery-driven electric machine.
- The system complexity (and cost) increase heavily depends on the hybrid architecture.
- Every hybrid system architecture has specific requirements in terms of electrical machine, power electronics and battery size/type.
- The advantages and the additional features gained through the adoption of hybrid architecture have the downside of creating issues in terms of integration (hardware and software).



Strictly confidential. Contents in this document are property of Kineton s.r.l. 2018



P0 • Easy system integration (but FEAD design shall be addressed)

- EM may (in some architectures) require liquid cooling
- Power electronics often integrated in the EM
- Reduced battery size
- Electric propulsion not available
- Some engine load shift (limited by EM power/torque and reduced battery size)

- Torque sustain during transients
- Limited energy recuperation
- 48V bus (no HV system requirements)
- Increased fuel efficiency (around 10%)

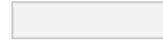


Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018

P1

- Higher torque than P0 (no belt)
- High impact on powertrain design
- Electric machines requires liquid cooling
- More complex power electronics
- Reduced battery size
- Electric propulsion not available
- Improved engine load shift respect to P0
- Improved torque fill during transients

- Limited energy recuperation
- 48V bus (no HV system requirements)
- Increased fuel efficiency respect to P0.
- No more investment on this architecture



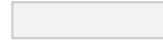
Confidential. All contents in this document are property of Kineton s.r.l. 2018



P2

- Increased energy recuperation potential (ICE can be decoupled from the wheels through the clutch)
- Electric creep/drive and e-coasting are possible
- High integration cost
- Larger battery is required to provide electric propulsion
- Starter cannot be eliminated to perform start&stop

- (but can be combined with BSG to obtain P0+P2)
- System efficiency is limited by the transmission between the EM and the wheels (both propulsion and recuperation)



Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018



- Highest energy recuperation potential
- Electric driving mode possible
- **In P4 configuration, the electric machine is not**

on the transmission drive axle, e-AWD is possible.

- Battery, motor, power electronics are larger and more expensive
- Fuel consumption improvement up to 25% on standard driving cycle (higher during urban driving).
- Starter is required to start the ICE (if not coupled with P0)
- A new design is required (no retrofit).

architecture • The EM is integrated into the transmission (on the even gears shaft) • Easy system integration (same ICE, main differences are in the transmission) • Electric driving mode possible

- Starter is required to start the ICE (if not coupled with P0)

Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018



P2.5

- Advantages in terms of system efficiency similar to P3

Strictly confidential. All contents in this document are property of Kineton s.r.l. 2018

